Introduction

In statistical quality control, quality means being fit for use and conformance to requirements. In broader terms, quality is the extent to which a product, service or process complies to a set of standards or benchmarks with low variability (variation). Quality is inversely proportional to variability; improvement of quality means reduction of variability in a process or product, and vice versa. Excessive variability in a process performance often results n waste.

Every product possesses certain elements that jointly describe what the users or consumers think of as quality. These are called quality characteristics or critical–to–quality (CTQ) characteristics. These characteristics maybe of several types:

a. **Physical:** length, weight, voltage, viscosity

b. **Sensory:** taste, appearance, colour

c. **Time orientation:** reliability, durability, serviceability,

Since variability can only be described in statistical terms, statistical methods play a central role in quality improvement efforts. In the application of statistical methods to quality engineering, data on quality characteristics can be classified as either attributes or variables data. Variables data are usually continuous measurements, such as length, voltage, or viscosity. Attributes data, on the other hand, are usually discrete data, often taking the form of counts, such as the number of loan applications, etc.

Statistical Methods for Quality Control and Improvement

Statistics is a collection of techniques useful for making decisions about a process or population based on the analysis of information contained in a sample from that population. Statistical methods play important role in quality control (QC) and improvement, by providing the principal means by which the product is sampled, tested and evaluated; information in those data are used to control and improve the manufacturing process.

Process Control

The following are some ways to control a process:

1. Through experience; this takes a lot of time,

2. Intuition, this is very risky,

3. By assuming that the process is in order; this may lead to rude awakening,

All these play important parts. They should be used as a support to a scientific approach in achieving and maintaining scientific control charts.

All these play important parts. They should be used as a support to a scientific approach in achieving and maintaining scientific control charts. The purpose of control is to identify and correct for the assignable causes as they occur and so to keep variation in the process within its natural limit. In doing so, the control chart is used to test whether the data represent random variations from stable sources, and if so, help to indicate the nature of sources for non–randomness.

Uses of Control Charts

1. **Judgment:** making judgment about the process such as establishing whether the process was in in a state of control in a given time,

2. **Maintaining Control:** it may be used in an ongoing effort to maintain the centering and spread of the process, and

3. It can be used to detect error for product change.

Definition of Terms

1. **Quality Control (QC):** the process of ensuring that specified standards (specifications) are maintained in every product or service; this can be physical, chemical, biological or any other standard.

2. **Quality Control Techniques:** the series of tests that are carried out separately to ensure that the standards are met.

3. **Statistical Quality Control:** the use of statistical methodology for improving or controlling production, product specifications, or inspection stages of any manufacturing process. It involves sampling and analysis of the results in order to arrive at the objective decision.

At the commencement of any manufacturing process, the specification(s) to be met by the product to be manufactured is/are usually set or made. Since any manufacturing process is subject to variation, every effort is made during the production (processing) of the material to meet the required specification (quality). Also at the end of the production, the produced items are inspected to see whether they conform to the required quality before being passed to the consumers.

Types of Deviations in Quality Control

a. Deviation within a unit

b. Deviation between units

c. Deviation over time

These deviations can simply be classified into two major groups, which are

- Random or chance variations: these occur naturally in every process and cannot be avoided

- Variations due to assignable causes like quantity of raw materials, operators, temperament, physical well–being of the machine, tool wears, electrical frustration, environment, newly employed staff, ect.

Four Sources of Assignable Causes (4m's)

a. **Man:** error from operators,

b. **Machine:** condition of the manufacturing machine

c. **Material:** raw materials used in production industries are mostly products of other companies, and if not in the best condition, they may cause variation

d. **Method:** this is the interplay of man–machine inflow and outflow of material. It is usually the major source of deviation in the product.

Measuring Quality Characteristics: This can be done using these measuring scales:

1. **Nominal Scale:** the measurement of nominal characteristics determines whether the characteristics of the object are the same as those of a particular realization or differ from it.

2. **Ordinal Scale:** here, the outcomes of a quality characteristics are in natural increasing or decreasing order, such as smaller or bigger, better or worse.

3. **Cardinal or Metric Scale:** this measures in numerals. The three different types here are interval, proportional and absolute scales.

Important Discrete Statistical Distributions

Several discrete probability distributions arise frequently in statistical quality control, such as the binomial, Poisson, and hypergeometric distributions.

The Binomial Distribution

Assuming we have a process consisting of a sequence of n independent trials, in which the outcome of each trial does not depend in any way on the outcomes of previous trials. When the outcome of each trial is either a "success" or a "failure," the trials are called Bernoulli trials. If we represent the probability of success on any trial with p, and p is constant, then the number of successes x in n Bernoulli trials has the binomial distribution with parameters n and p, defined as:

$$P(x)=\binom{n}{x}p^x(1-p)^{n-x}=\binom{n}{x}p^xq^{n-x}$$

$$n\geq 0, 0<p<1,\ q=1-p \text{ and } x=0,1,2,...,n$$

The mean of the binomial distribution is $\mu=np$, while the variance is $\sigma^2=np(1-p)$.

The binomial distribution is used frequently in quality engineering. It is the appropriate probability model for sampling from an infinitely large population, where p represents the fraction of defective or nonconforming items in the population. In these applications, x usually represents the number of nonconforming items found in a random sample of n items.

Example 1

A random sample of 5 was selected from a steady string of production of a newspaper press, and the fraction defective is 10%:

a. what is the probability that there is one defective in the sample?
b. what is the probability that there is one or fewer defects in the sample?
c. what is the probability that there are two or more defects in the sample?

Solution

$n=5, p=0.1=10\%,\ q=1-0.1=0.9$

a. $P(x=1)=\binom{5}{1}(0.1)^1(0.9)^{5-1}=0.328$

b. $x=1$ or less implies $x=1$ or $x=0$

$$=P(x=1)+P(x=0)$$
$$=\binom{5}{1}(0.1)^1(0.9)^{5-1}+\binom{5}{0}(0.1)^0(0.9)^{5-0}$$
$$=0.328+0.590=0.918$$

c. $P(x\geq 2)=P(x=2)+P(x=3)+P(x=4)+P(x=5)$

$$P(x=2)=\binom{5}{2}(0.1)^2(0.9)^{5-2}=0.0729$$

$$P(x \le 2) = \frac{\ell^{-4}4^0}{0!} + \frac{\ell^{-4}4^1}{1!} + \frac{\ell^{-4}4^2}{2!}$$

$$= \ell^{-4}(1+4+8) = 0.238$$

Exercise Two

1. Suppose that a secretary commits two typos on average per page. What is the probability that she commits:

a. no typos,
b. less than two typos
c. four or more typos on the next page typed

The Hypergeometric Distribution

Suppose that there is a finite population consisting of N items. Some number say, $D(D \le N)$ of these items fall into a class of interest. A random sample of n items is selected from the population without replacement, and the number of items in the sample that fall into the class of interest say, x is observed. Then x is a hypergeometric random variable with the probability distribution defined as follows:

$$P(x) = \frac{\binom{D}{x}\binom{N-D}{n-x}}{\binom{N}{n}}$$

$$x = 0,1,2,\ldots,\min(n, D)$$

Mean and variance of the hypergeometric distribution are $\mu = \frac{nD}{N}$, and $\sigma^2 = \frac{nD}{N}\left(1 - \frac{D}{N}\right)\left(\frac{N-n}{N-1}\right)$

The hypergeometric distribution is the appropriate probability model for selecting a random sample of n items without replacement from a lot of N items of which D are nonconforming or defective. By a random sample, we mean a sample that has been selected in such a way that all possible samples have an equal chance of being chosen. In these applications, x usually represents the number of nonconforming items found in the sample.

Example 3

Suppose that a lot contains 100 items, 5 of which do not conform to requirements. If 10 items are selected at random without replacement, then the probability of finding one or fewer nonconforming items in the sample is:

$$P(x \le 1) = P(x = 0) + P(x = 1)$$

$$= \frac{\binom{5}{0}\binom{95}{10}}{\binom{100}{10}} + \frac{\binom{5}{1}\binom{95}{9}}{\binom{100}{10}} = 0.92314$$

$$P(x=3)=\binom{5}{3}(0.1)^3(0.9)^{5-3}=0.0081$$

$$P(x=4)=\binom{5}{4}(0.1)^4(0.9)^{5-4}=0.00045$$

$$P(x=5)=\binom{5}{5}(0.1)^5(0.9)^{5-5}=0.00001$$

$$\therefore P(x\geq 2)=0.0729+0.0081+0.00045+0.00001=0.08146$$

Exercise One

1. Suppose that it is known that 10% of the components produced by a certain machine are defective. If four components are randomly selected, find the probability that:
   a. exactly one component is defective,
   b. at least one component is defective,
   c. two or three components are defective

2. A random sample of 15 loaves of bread was selected from 10 sets of baked bread in a company. If 0.1 is the fraction of nonconforming bread, find the probability of getting:
   a. one nonconforming bread,
   b. two or three nonconforming breads,
   c. more than three nonconforming breads.

The Poisson Distribution

The Poisson distribution is a useful discrete distribution in statistical quality control. It is defined as:

$$P(x)=\frac{\ell^{-\lambda}\lambda^x}{x!}$$

$\lambda>0$ and $x=0,1,2,...$

The mean and variance of the Poisson distribution are $\mu=\sigma^2=\lambda$.

A typical application of the Poisson distribution in quality control is as a model of the number of defects or nonconformities that occur in a unit of product. In fact, any random phenomenon that occurs on a per-unit (or per unit area, per unit volume, per unit time, etc.) basis is often well approximated by the Poisson distribution.

Example 2

Suppose that the number of wire–bonding defects per unit that occur in a semiconductor device is Poisson distributed with parameter $\lambda=4$. Then the probability that a randomly selected semiconductor device will contain two or fewer wire–bonding defects is $P(x\leq 2)$ and is given as follows:

$$P(x\leq 2)=P(x=0)+P(x=1)+P(x=2)$$

$$P(x\leq 2)=\frac{\ell^{-4}4^x}{x!};\ x=0,1,2$$

Example 4
A shipment of 20 TV sets contains 5 defective items. A sample of 4 was selected from the lot without replacement. Find the probability of getting 2 defective items out of the 4 selected.

Solution
$D=5,\ x=2,\ N=20,\ n=4$

$$P(x=2)=\frac{\binom{5}{2}\binom{20-5}{4-2}}{\binom{20}{4}}=\frac{\binom{5}{2}\binom{15}{2}}{\binom{20}{4}}=0.2167$$

Example 5
A shipment of 15 cameras contains 4 bad and 11 good ones. If a sample of six cameras is selected without replacement one after the other. Find:
a. the probability of getting three bad ones in six trials
b. the probability of getting two bad ones in six trials

Solution
a. $P(x=3)$: $D=4,\ x=3,\ N=15,\ n=6$

$$P(x=3)=\frac{\binom{4}{3}\binom{15-4}{6-3}}{\binom{15}{6}}=\frac{\binom{4}{3}\binom{11}{3}}{\binom{15}{6}}=0.1319$$

b. $P(x=2)$: $D=4,\ x=2,\ N=15,\ n=6$

$$P(x=3)=\frac{\binom{4}{2}\binom{15-4}{6-2}}{\binom{15}{6}}=\frac{\binom{4}{2}\binom{11}{4}}{\binom{15}{6}}=0.3956$$

A double-sampling plan is a procedure in which, under certain circumstances, a second sample is required before the lot can be sentenced. A double-sampling plan is defined by four parameters:

n_1 = sample size on the first sample
c_1 = acceptance number of the first sample
n_2 = sample size on the second sample
c_2 = acceptance number for both samples

As an example, suppose $n_1 = 50$, $c_1 = 1$, $n_2 = 100$ and $c_2 = 3$. Thus, a random sample of $n_1 = 50$ items is selected from the lot, and the number of defectives in the sample, d_1, is observed. If $d_1 \le c_1 = 1$, the lot is accepted on the first sample. If $d_1 > c_2 = 3$, the lot is rejected on the first sample. If $c_1 > d_1 \le c_2$, a second random sample of size $n_2 = 100$ is drawn from the lot, and the number of defectives in this second sample, d_2, is observed. Now the combined number of observed defectives from both the first and second sample, $d_1 + d_2$, is used to determine the lot sentence. If $d_1 + d_2 \le c_2 = 3$, the lot is accepted. However, if $d_1 + d_2 > c_2 = 3$, the lot is rejected.

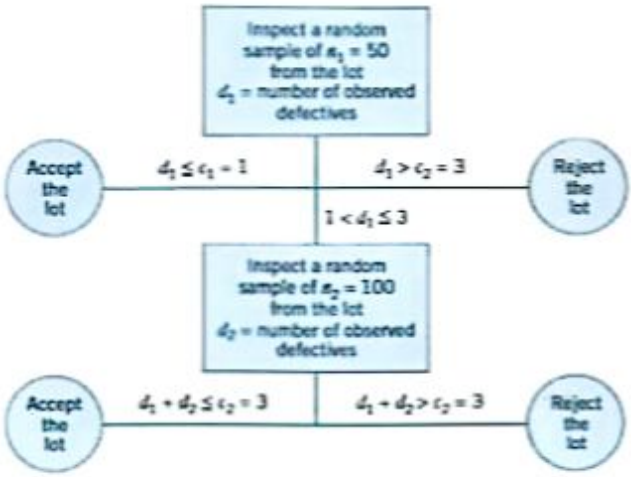


Figure 2: Illustration of the double sampling plan, $n_1 = 50$, $c_1 = 1$, $n_2 = 100$ and $c_2 = 3$.

Further illustration is as follows:
Given that $N = 1000$, $n_1 = 50$, $c_1 = 1$, $n_2 = 100$ and $c_2 = 3$, according to the double sampling plan

a. Each batch contains 1000 items or products,
b. Pick out a sample of 50 items and inspect them,
c. If the no of defective items is less than or equal to 1, $(d_1 \le 1)$ accept the batch of 1000 and discontinue inspection,
d. If $d_1 > 3$, reject the whole batch of 1000,
e. If $d_1 > 1$, but less than 3, $(c_1 < d_1 \le c_2)$, then take a second sample of 100 from the batch, remaining 950,

f. If the total number of defectives in 50+100=150 items is less than or equal to 3 $(d_1 + d_2 \le c_2 = 3)$, then accept the batch of 1000, otherwise reject it.

A double sampling plan is usually more effective because it requires less inspection compared to single sampling plan.

Example 7

A random sample of 100 items was selected, and the batch is accepted if it contains fewer than 2 defectives and rejected if it contains more than 3 defectives. If there are 2 or more defectives in the sample, a second sample of size 50 is selected, and the batch is accepted if the pooled sample contains fewer than 4 defectives, otherwise it is rejected. Calculate the probability of accepting batches if the process average is 0.01.

Solution

$n_1 = 100$, $c_1 = 0$ or 1, $n_2 = 50$ and $c_2 \le 3$

$P(accept/p) = P(d_1 = 0) \text{ or } P(d_1 = 1) + P(d_1 = 2) \times P(d_2 = 0) + P(d_1 = 2) \times P(d_2 = 1) +$
$P(d_1 = 3) \times P(d_2 = 0)$.

The Poisson approximation to the binomial will be used to solve this problem as follows

$$\frac{\lambda^x \ell^{-\lambda}}{x!} = \frac{(np)^x \ell^{-(np)}}{x!}$$

$\lambda = np$; p=0.001. $n_1 = 100$, $n_2 = 50$; $n_1 p = 100 \times 0.01 = 1$; $n_2 p = 50 \times 0.01 = 0.5$.

The required probability is given by:

$$\left(\frac{1^0 \ell^{-1}}{0!} + \frac{1^1 \ell^{-1}}{1!}\right) + \left(\frac{1^2 \ell^{-1}}{2!} \times \frac{0.5^0 \ell^{-0.5}}{0!}\right) + \left(\frac{1^2 \ell^{-1}}{2!} \times \frac{0.5^1 \ell^{-0.5}}{1!}\right) + \left(\frac{1^3 \ell^{-1}}{3!} \times \frac{0.5^0 \ell^{-0.5}}{0!}\right)$$

$0.7358 + 0.1116 + 0.0558 + 0.0372 = 0.9404$

Multiple Sampling Plans

A multiple sampling plan is an extension of double-sampling in that more than two samples can be required to sentence a lot. An example of a multiple sampling plan with five stages follows.

Cumulative Sample Size	Acceptance Number	Rejection Number
20	0	3
40	1	4
60	3	5
80	5	7
100	8	9

This plan will operate as follows: if, at the completion of any stage of sampling, the number of defective items is less than or equal to the acceptance number, the lot is accepted. If, during any

stage, the number of defective items equals or exceeds the rejection number, the lot is rejected; otherwise the next sample is taken. The multiple-sampling procedure continues until the fifth sample is taken, at which time a lot disposition decision must be made. The first sample is usually inspected 100%, although subsequent samples are usually subject to curtailment.

Sequential Sampling Plans

Sequential sampling is an extension of the double sampling and multiple sampling concept. In sequential sampling, we take a sequence of samples from the lot and allow the number of samples to be determined entirely by the results of the sampling process. In practice, sequential sampling can theoretically continue indefinitely, until the lot is inspected 100%. In practice, sequential sampling plans are usually truncated after the number inspected is equal to three times the number that would have been inspected using a corresponding single-sampling plan. If the sample size selected at each stage is greater than one, the process is usually called *group* sequential-sampling. If the sample size inspected at each stage is one, the procedure is usually called item-by-item sequential-sampling.

Process Control (PC)

This deals with control of the quality of goods manufactured during the process of production. PC also controls the quality of goods to be produced. Its main aim is to determine whether the production processes are going on as designed, thereby turning out products of required standards. This is usually achieved via the use of control charts. In the PC, an attempt is made to find out if a process is within control. A process is said to be in control when its results in the future will be the same as they have been in the past. Statistically, the process is under control if the sample mean is approximately equal to the population mean, that is if $\bar{x} = \mu$, while it is out of control if there is a change in process mean (sample mean) from population mean. Also, if the average falls outside the control limit, it shows strongly that the process is out of control. The theory of acceptance sampling deals with the problem of rating the quality of finite population of produced units after completion of the manufacturing processes, but the process control intends to represent the basic idea of statistical process control which tries to detect quality deficiency during manufacturing process. Since in nature and in manufacturing, no two items are ever exactly alike, the ability to measure the variations in the objects is necessary before it can be controlled.

Categories of Variations

a. **Within–Piece Variation:** variation within one piece of object, such as degrees of smoothness of a ball bearing.
b. **Piece–Piece Variation:** variation between objects produced at the same time, such as difference in the intensity of light bulbs of the same rating produced at the same time from the same machine.
c. **Time–Time Variation:** this happens when goods produced at one time of the day vary from those produced at other times of the day.

There could also be two other lines, which are:

i. upper warning limit (UWL),
ii. lower warning limit (LWL)

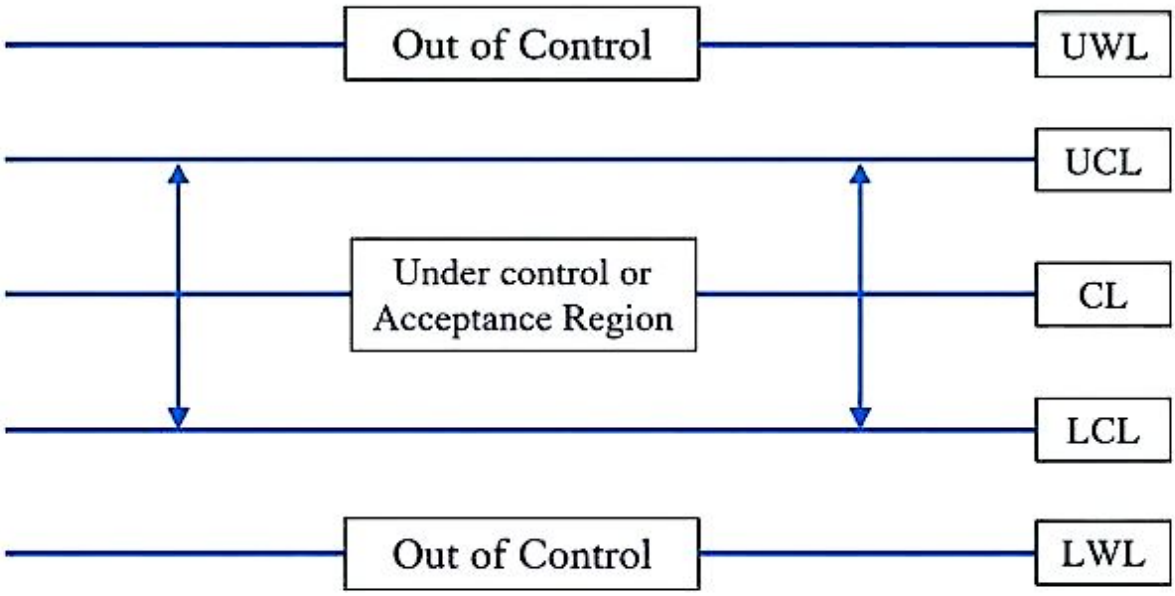


Figure 3: Graph showing the lines of a control chart

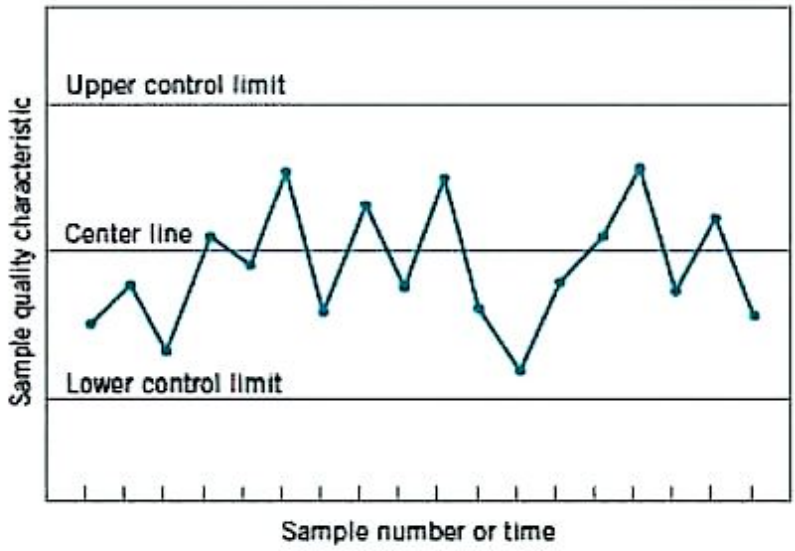


Figure 4: *A typical Control Chart*

Chance and Assignable Causes of Quality Variation

Chance cause of quality variation is the amount of inherent or natural variability that always exist in the production process. This natural variability or "background noise" is the cumulative effect of many small, essentially unavoidable causes. In other words, the chance causes are an inherent part of the process. A process that is operating with only chance causes of variation present is said to be in statistical control.

Assignable causes of quality variation are the variations which usually arise from improperly controlled or adjusted machines, operator errors, or defective raw materials, which are generally large in comparison to chance causes of variation. They usually represent an unacceptable level of process performance. A process that is operating in the presence of assignable causes is said to be an out-of-control process.

The Control Chart

This is a graphical representation of mathematical model used to monitor a process in order to detect changes in the parameters of the process.

Uses of Control Charts

a. It serves as a function of alerting those responsible to the possibility that the process is not working as expected.
b. It continuously monitors the manufacturing process to know when it is under statistical control.
c. It is a powerful tool for controlling and stabilizing the process at the required stability level.
d. It ensures the uniformity of manufactured items.
e. It may also be used for judging whether a control has been attained or not.
f. It is also a device for attaining control.
g. Control charts may also be used to estimate the parameters of a production process, and, through this information, to determine process capability.
h. The control chart may also provide information useful in improving the process.
i. The control chart is an effective tool in reducing variability as much as possible, since the eventual goal of statistical process control is the elimination of variability in the process.

Control chart makes use of measurement of a particular dimension, performance characteristics, or other continuous scaled variable such as diameter or it may be based on attribute. If variation in the data studied is in agreement with statistical pattern, then the condition producing the variation is under control or vice versa. Therefore, the process of control chart analysis is said to be in statistical control when the process is governed wholly by a random cause which are large in magnitude.

The control chart has three major lines, which are:

a. center or central limit (CL) line,
b. upper control line (UCL) or upper allowable or action limit,
c. lower control line (LCL) or lower allowable or action limit,